

Question + Video

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graph TD; A[Question + Video] --> B[Reconstruct once ❄️  
11.3 s / scene →  
0.7 s / question]; A --> C[Pose-keyed caches  
render 0.12 s;  
repeats are free]; B --> D[Gated, bounded search 🔥  
71% exit in 9.6 s;  
b=3, keep 2, d≤3]; C --> E[Decode-free control  
prefill logits only;  
fused head −4.0 s]; D --> F[Answer: 24.5 s / 1.1 kJ avg.]; E --> F;
```

**Reconstruct
once ❄️**

11.3 s / scene →
0.7 s / question

**Pose-keyed
caches**

render 0.12 s;
repeats are free

**Gated, bounded
search 🔥**

71% exit in 9.6 s;
b=3, keep 2, $d \leq 3$

**Decode-free
control**

prefill logits only;
fused head −4.0 s

Answer: 24.5 s / 1.1 kJ avg.